Unit 4

DECIMALS AND PERCENTAGES

4.1 DECIMALS

Decimal is a number that contains fractional part, such as: 0.4, 6.5, 17.23 are all decimals. We also know that the number of digits at the right hand side of a decimal point determines the number of decimal places. For example 7.6, 4.8, 0.2, 0.9 are all decimal fractions up to one decimal place.

6.37, 4.95, 0.12, 9.08 are all decimal fractions up to two decimal places.

Similarly a fraction which has three digits after decimal point is called a decimal fraction up to three decimal places. 1.002, 4.036, 5.123 are all decimal fractions up to three decimal places.

In decimal fraction, the place values of the digits increase 10 times on moving from right to left. Conversely reduce to $\frac{1}{10}$ th on proceeding from left to right.

Place value of 111.111, 222.222, 333.333 and 888.888 are given below:

Number	Hundreds	Tens	Ones	Tenths	Hundredths	Thousandths
111.111	100	10	1	$\frac{1}{10} = 0.1$	$\frac{1}{100} = 0.01$	$\frac{1}{1000} = 0.001$
222.222	200	20	2	$\frac{2}{10} = 0.2$	$\frac{2}{100} = 0.02$	$\frac{2}{1000} = 0.002$
333.333	300	30	3	$\frac{3}{10} = 0.3$	$\frac{3}{100} = 0.03$	$\frac{3}{1000} = 0.003$
888.888	800	80	8	$\frac{8}{10} = 0.8$	$\frac{8}{100} = 0.08$	$\frac{8}{1000} = 0.008$

Teacher's Note

Teacher should explain the students, when the digit moves one step to the right, the place value becomes $\frac{1}{10}$ (one-tenth).

Let us consider the place value of the position of each digit in 765.984

Hundreds	Tens	Ones	Decimal	Tenths	Hundredths	Thousandths
7	6	5	.	9	8	4

The decimal point separates the whole number part from the fractional part. If a number consists of only a decimal part, then we take zero as whole number.

For Example:

.45 = 0.45, .06 = 0.06 and .007 = 0.007

Again if a number consists of only a whole number part and we want to describe it in decimals, we take zero as decimal. For example 36 = 36.0 upto one decimal place or 36.00 upto two decimal places.

4.1.1 Add and subtract decimals

In previous class we have learnt about decimals, conversion between fractions and decimals and basic operations on decimals.

Now we learn addition and subtraction of decimals.

Let us consider the following examples:

Example 1. Add the following:

(i) 20.25 + 7.52 (ii) 234.452 + 23.23

(iii) 109.25 + 7.589 (iv) 608.56 + 23.068

Solution: (i) 20.25 + 7.52

For adding these decimal fractions, write the given fractions in vertical form in such a way, that decimal points of both addends will be placed right below each other. Then add by using rules as we have already learned.

Teacher's Note Teacher should help the students to put the decimal point in the correct place.

(i)
```
  20.25
+  7.52   means
  27.77
```

	Tens	Ones	Decimal	Tenths	Hundredths
	2	0	.	2	5
+	0	7	.	5	2
	2	7	.	7	7

Write zero (0) where there is no digit like in the given example.

(ii)
```
  234.452
+  23.230
  257.682
```

(iii)
```
  109.250
+   7.589
  116.839
```

(iv)
```
  608.560
+  23.068
  631.628
```

Example 2. Subtract the following:

(i) 58.75 – 17.50 (ii) 782.65 – 293.562

(iii) 422.785 – 206.5 (iv) 845. 506 – 458.068

Solution: (i) 58.75 – 17.50

For subtracting these decimal fractions, write the given fractions in vertical form in such a way that decimal points of both the numbers will be right below each other. Then subtract by using rules as we have already learned.

(i)
```
  58.75
– 17.50 means
  41.25
```

	Tens	Ones	Decimal	Tenths	Hundredths
	5	8	.	7	5
–	1	7	.	5	0
	4	1	.	2	5

Write zero (0) where there is no digit like in the given example.

(ii)
```
  782.650
– 293.562
  489.088
```

(iii)
```
  422.785
– 206.500
  216.285
```

(iv)
```
  845.506
– 458.068
  387.438
```

EXERCISE 4.1

A. Add the following:

(1) 22.32 + 6.46 (2) 4.567 + 36.4 (3) 75.05 + 24.62
(4) 257.003 + 0.25 (5) 40.123 + 7.32 (6) 45.005 + 52.47
(7) 345.38 + 786.46 (8) 674.567 + 36.48
(9) 45.75 + 54.69 (10) 287.099 + 8.258
(11) 45.468 + 277.358 (12) 35.69 + 875.875

B. Subtract the following:

(1) 25.52 – 6.3 (2) 74.567 – 33.402
(3) 75.75 – 24.62 (4) 257.003 – 0.25
(5) 49.123 – 7.02 (6) 757.785 – 152.005
(7) 786.46 – 345.38 (8) 674.567 – 36.48
(9) 85.75 – 54.65 (10) 287.099 – 174.055
(11) 845.468 – 234.358 (12) 935.69 – 805.365

4.1.2 Recognize like and unlike decimals

In previous class we have learnt fractions with same denominators. For example, $\frac{2}{7}$, $\frac{3}{7}$, $\frac{4}{7}$, $\frac{5}{7}$. Such fractions which have same denominators are known as like fractions. Similarly fractions with different denominators are known as unlike fractions for example $\frac{3}{5}$, $\frac{2}{7}$ and $\frac{1}{2}$ etc.

Decimal fractions with same number of decimal places like 4.5, 6.3, 56.7 are some examples of **like decimals**.

The decimals like 34.0, 567.24, 234.7802 are some **unlike decimals**.

Example 1: Recognize decimals of the 2-decimal places of the following:

(i) 6.3, 0.25, 25.52, 643.2, 342.81, 14.025, 67.9 and 8.01

Solution:

0.25, 25.52, 342.81 and 8.01 are decimals with two decimal places.

4.1.3 Multiply decimals by 10, 100 and 1000

Consider the following examples.

Example 1. Find the product.

(i) 2.23 x 10 (ii) 2.23 x 100 (iii) 2.23 x 1000

Solution:

(i) 2.23 x 10 = 2.230 = 22.30 = **22.3**

What happen to value of 2.23

Multiply a decimal by 10, then the value of the decimals increases ten times or the decimal point moves one place to the right.

(ii) 2.23 x 100 = 2.2300 = 223.00 = **223.**

Multiply a decimal by 100 then the value of the decimals increases hundred times or the decimal point moves two places to the right.

(iii) 2.23 x 1000 = 2.23000 = 2230.00 = **2230.**

Multiply a decimal by 1000, then the value of the decimals increases one thousand times or the decimal point moves three places to the right.

EXERCISE 4.2

A. Separate the decimals of one, two and three decimal places from the following.

0.08, 2.123, 34.25, 0.6, 3.36, 52.302, 38.66 and 62.1.

B. Find the product of the following by using the rules of multiplication:

(1) 0.175 x 10	(2) 0.175 x 100	(3) 0.175 x 1000
(4) 35.058 x 10	(5) 35.058 x 100	(6) 35.058 x 1000
(7) 8.15 x 10	(8) 8.15 x 100	(9) 8.15 x 1000
(10) 324.423 x 10	(11) 324.423 x 100	(12) 324.423 x 1000
(13) 0.0067 x 10	(14) 0.0067 x 100	(15) 0.0067 x 1000

4.1.4 Divide decimals by 10, 100 and 1000

Rules:

1. Divide a decimals by10, then the value of the decimals. Decreases ten times or the decimal point moves one place to the left

$$\boxed{1\,5.34 \div 10 = 1.534}$$

(What happened to value of 15.35?)

2. Divide a decimals by100, then the value of the decimals decreases hundred times or the decimal point moves two places to the left.

$$\boxed{15.34 \div 100 = 0.1534}$$

3. Divide a decimals by1000, then the value of the decimals decreases one thousand times or the decimal point moves three places to the left.

$$\boxed{015.34 \div 1000 = 0.01534}$$

Example 1: **Solve by using the above stated rules:**

(1) 0.0175 ÷ 10 (2) 0.0175 ÷ 100

(3) 0.0175 ÷ 1000

Solution:

The value of decimals decreases accordingly 10, 100 and 1000

(1)	(2)	(3)
0.0175 ÷ 10	0.0175 ÷ 100	0.0175 ÷ 100
0.0175 ÷ 10	00.0175 ÷ 100	000.0175 ÷ 1000
= **0.00175**	= **0.000175**	= **0.000175**

EXERCISE 4.3

A. Solve the following by using the rules of division.

(1) 6.675 ÷ 10	(2) 6.675 ÷ 100	(3) 6.675 ÷ 1000
(4) 35.89 ÷ 10	(5) 35.89 ÷ 100	(6) 35.89 ÷ 1000
(7) 815.4 ÷ 10	(8) 815.4 ÷ 100	(9) 815.4 ÷ 1000
(10) 0.085 ÷ 10	(11) 0.085 ÷ 100	(12) 0.085 ÷ 1000

4.1.5 Multiply a decimal with a whole number

We consider the following examples.

Example 1. **Find the product:**

(i) 0.231 x 2 (ii) 4.4 x 4 (iii) 2.3 x 24

Solution:

(i) 0.231 x 2 = $\begin{array}{r} \mathbf{0.231} \\ \times\ \mathbf{2} \\ \hline \mathbf{0.462} \\ \hline \end{array}$

(ii) 4.4 x 4 = $\begin{array}{r} \mathbf{4.4} \\ \times\ \mathbf{4} \\ \hline \mathbf{17.6} \\ \hline \end{array}$

(iii) 2.3 x 24

Solution:

$$\begin{array}{r} 2.3 \\ \times\ 24 \\ \hline 92 \\ 460 \\ \hline 55.2 \\ \hline \end{array}$$

Step 1: Neglect the decimal point. Multiply 3 by 4, We get 12, write 2 at first place and carry 1. Multiply 2 by 4, we get 8, add 1 in 8, we get 9, write in second place, we get 92.

Step 2: Write 0 in ones place. Multiply 3 by 2, we get 6, write 6 after zero. Now multiply 2 by 2, we get 4.

Step 3: Add all the digits of ones and tens, we get 552.

Step 4: Count decimal places in given numbers to be multiplied.

Step 5: Put a decimal point after one digit, count from right digit. We get 55.2

EXERCISE 4.4

Find the product:

(1) 6.5 x 5	(2) 3.5 x 6	(3) 0.65 x 5
(4) 0.65 x 6	(5) 0.35 x 8	(6) 4.25 x 7
(7) 4.25 x 8	(8) 4.25 x 9	(9) 0.382 x 5
(10) 0.394 x 8	(11) 24.58 x 7	(12) 53.69 x 9
(13) 4.25 x 27	(14) 0.265 x 36	(15) 2.785 x 435
(16) 4.25 x 162	(17) 47.326 x 348	(18) 58.967 x 564

4.1.6 Divide a decimal with a whole number.

Consider the following examples:

Example 1. Solve: 0.9 ÷ 3

Solution: 0.9 ÷ 3 = 9 Tenths ÷ 3

= (9÷3) Tenths = 3 Tenths

= 0.3

$$3\,\overline{)\,0.9\,}\,(0.3 \qquad \begin{array}{r} -9 \\ \hline 0 \end{array}$$

Example 2. Solve: 0.84 ÷ 4

Solution: 0.84 ÷ 4 = (0.80 + 0.04) ÷ 4

= (8 tenths ÷ 4) + (4 hundredths ÷ 4)

= 2 tenths + 1 hundredth

= 20 hundredths + 1 hundredth = 21 hundredths

= 0.21

$$4\,\overline{)\,0.84\,}\,(0.21 \qquad \begin{array}{r} -8 \\ \hline 4 \\ -4 \\ \hline 0 \end{array}$$

Teacher's Note Teacher may ask students to see and note how the decimals are lined up in the original number and in the answer (quotient).

Example 3. **Solve: 25.85 ÷ 5**

Solution: **(25 + 0.8 + 0.05) ÷ 5**

25.85 ÷ 5 = (25 ÷ 5) + (8 Tenths ÷ 5) + (5 Hundredths ÷ 5)

$$= (5) + (0.8 \div 5) + (0.05 \div 5)$$
$$= 5 + 0.16 + 0.01$$
$$= 5.17$$

OR

$$\begin{array}{r|l} 5 & 25.85 \; (5.17 \\ & -25 \\ \hline & 8 \\ & -5 \\ \hline & 35 \\ & -35 \\ \hline & 0 \end{array}$$

Verify

5.17	Quotient
× 5	Divisor
25.85	Dividend

So, 25.85 ÷ 5 = 5.17

EXERCISE 4.5

A. Solve:

(1) 0.65 ÷ 5 (2) 6.5 ÷ 5 (3) .065 ÷ 5
(4) 3.6 ÷ 6 (5) 0.64 ÷ 8 (6) 4.27÷ 7
(7) 9.6 ÷ 8 (8) 44.1 ÷ 9 (9) 0.385 ÷ 5
(10) 8.394 ÷ 6 (11) 39.851 ÷ 7 (12) 87.03 ÷ 9

B. Solve up to 3-decimal places:

(1) 40.25 ÷ 4 (2) 89.64 ÷ 6 (3) 98.58 ÷ 3
(4) 24.25 ÷ 10 (5) 0.265 ÷ 20 (6) 2.785 ÷ 25
(7) 3.16 ÷ 2.31 (8) 0.555 ÷ 5.5 (9) 14.04 ÷ 12.4

4.1.7 Multiply a decimal by tenths and hundredths only

We have already learnt:

One-Tenth = 0.1 and One-Hundredth = 0.01

Now consider the following examples.

Example 1. Find the product:

(i) 2.23 x 0.1 (ii) 2.23 x 0.01

(iii) 2.23 x 0.5 (iv) 2.23 x 0.07

Solution:

(i)
$$\begin{array}{r} 2.23 \\ \times\ 0.1 \\ \hline 0.223 \\ \hline \end{array}$$

(ii)
$$\begin{array}{r} 2.23 \\ \times\ 0.01 \\ \hline 0.0223 \\ \hline \end{array}$$

(iii)
$$\begin{array}{r} 2.23 \\ \times\ 0.5 \\ \hline 1.115 \\ \hline \end{array}$$

(iv)
$$\begin{array}{r} 2.23 \\ \times\ 0.07 \\ \hline 0.1561 \\ \hline \end{array}$$

Example 2. Find the product of: 0.04 x 0.2

Solution:

$$0.04 \times 0.2 = 0.008 \rightarrow \underbrace{0.04}_{\text{2 decimal places}} \times \underbrace{0.2}_{\text{1 decimal place}} = \underbrace{0.008}_{\text{3 decimal places}} \text{ (product)}$$

in 0.04 we have 2 decimal places and in 0.2 we have one decimal place. When we write their product then add decimals and get 3 decimal places. Count from right up to 3 decimal places and if digits are less than 3 places, write zeros as we need, as shown in the example, we write 2 zeros.

Thus, we get 0.008

EXERCISE 4.6

Find the product of the following:

(1) 2.6 x 0.1	(2) 3.87 x 0.2	(3) 42.5 x 0.3
(4) 59.95 x 0.4	(5) 0.3 x 0.5	(6) 0.01 x 0.6
(7) 1.1 x 0.4	(8) 1.05 x 0.2	(9) 2.6 x 0.03
(10) 0.3 x 0.01	(11) 0.01 x 0.01	(12) 1.1 x 0.08
(13) 3.87 x 0.06	(14) 42.5 x 0.05	(15) 59.95 x 0.07

4.1.8 Multiply a decimal by a decimal (with three decimal places)

Consider the following examples:

Example 1. Find the product of: 0.25 x 0.056

Solution:

0.25 x 0.056 =

$$\begin{array}{r l} 0.056 & \longrightarrow \text{3 decimal places} \\ \times\ 0.25 & \longrightarrow \text{2 decimal places} \\ \hline 280 & \\ 112\times & \\ \hline 1400 & \\ \hline \end{array}$$

So, we get the product (3 + 2 = 5 decimals) **0.01400 = 0.014**

Or

$$\underbrace{0.25}_{\text{2 decimal place}} \times \underbrace{0.056}_{\text{3 decimal places}} = \underbrace{0.01400}_{\text{5 decimal places}} = 0.014$$

Teacher's Note Teacher should ask students to write the given number and the product in place value chart, then compare the values of both.

Example 2. **Solve 0.045 x 0.68**

Solution:

$$\underbrace{0.045}_{\text{3 decimal places}} \times \underbrace{0.68}_{\text{2 decimal places}} = \underbrace{0.03060}_{\text{5 decimal places}} = 0.0306$$

	0.0 4 5
x	0. 6 8
	3 6 0
	2 7 0 x
	3 0 6 0

Thus, 0.045 x 0.68 = 0.03060 = **0.0306**

4.1.9 Multiply a decimal by a decimal (in the same way as for whole numbers and then in the decimal point accordingly)

To multiply a decimal by a decimal in the same way as for whole numbers. Let us consider the following examples.

Example:

Find the product:

(i) 1.9 x 2.7 (ii) 28.5 x 1.25

Solutions:

(i) Number of decimal places in 1.9 is one.
Number of decimal places in 2.7 is one .
Number of decimal places in the product will be 1 + 1 = 2 .
Therefore, 1.9 x 2.7 = **5.13**

	1.9
x	2.7
	133
	38 x
	5.13

(ii) Number of decimal places in 28.5 is one and in 1.25 is two.
Number of decimal places in the product will be 1 + 2 = 3.
Therefore, 28.5 x 1.25 = **35.625**

	28.5
x	1.25
	1425
	570 x
	285 xx
	35.625

EXERCISE 4.7

A. Put the decimal point at the appropriate place in each answer.

(1) 46.7 x 1.2 = 5604

(2) 0.754 x 1.2 = 09048

(3) 3.57 x 1.2 = 4284

(4) 68.2 x 1.2 = 8184

(5) 0.402 x 1.2 = 04824

(6) 14.839 x 1.2 = 178068

(7) 361.4 x 3.54 = 1279356

(8) 219.241 x 2.5 = 5481025

(9) 402.58 x 2.3 = 925934

B. Solve:

(1) 0.28 x 0.4	(2) 0.45 x 0.5	(3) 0.2 x 0.023
(4) 0.28 x 0.05	(5) 0.065 x 0.25	(6) 0.005 x 0.12
(7) 0.002 x 0.08	(8) 0.105 x 0.09	(9) 0.31 x 0.052
(10) 0.22 x 0.057	(11) 0.25 x 0.05	(12) 0.755 x 0.14

C. Find the product:

(1) 1.4 x 2.5	(2) 1.05 x 2.6	(3) 3.8 x 0.7
(4) 0.3 x 2.01	(5) 4.45 x 1.8	(6) 8.84 x 1.5
(7) 6.24 x 7.1	(8) 5.25 x 4.4	(9) 7.05 x 2.6
(10) 9.06 x 5.5	(11) 25.08 x 8.5	(12) 9.2 x 7.85

4.1.10 Divide a decimal by a decimal (by converting decimals to fractions)

Let us consider the following examples:

Example 1. Solve: 2.48 ÷ 1.24

Solution:

$$2.48 \div 1.24 = \frac{248}{100} \div \frac{124}{100}$$

$$= \frac{248}{100} \times \frac{100}{124}$$

$$= \frac{248 \times \overset{1}{\cancel{100}}}{\underset{1}{\cancel{100}} \times 124} = \frac{248}{124}$$

$$= \frac{\overset{\overset{\overset{2}{\cancel{62}}}{\cancel{124}}}{\cancel{248}}}{\underset{\underset{\underset{1}{\cancel{31}}}{\cancel{62}}}{\cancel{124}}} = 2$$

Thus, 2.48 ÷ 1.24 = **2**

Example 2: Solve: 1.84 ÷ 2.3

Solution:

$$1.84 \div 2.3 = \frac{184}{100} \div \frac{23}{10}$$

$$= \frac{184}{100} \times \frac{10}{23}$$

$$= \frac{\overset{8}{\cancel{184}} \times \overset{1}{\cancel{10}}}{\underset{10}{\cancel{100}} \times \underset{1}{\cancel{23}}}$$

$$= \frac{8}{10} = 0.8$$

Thus, 1.84 ÷ 2.3 = **0.8**

Example 3: Solve: 6.25 ÷ 0.25

Solution: $6.25 \div 0.25 = \frac{625}{100} \div \frac{25}{100}$

$$= \frac{625}{100} \times \frac{100}{25} = \frac{625 \times \cancel{100}^{1}}{\cancel{100}_{1} \times 25}$$

$$= \frac{625}{25} = \frac{\cancel{625}^{25}}{\cancel{25}_{1}} = 25$$

Thus 6.25 ÷ 0.25 = **25**

4.1.11 Divide a decimal by a decimal using direct division by moving decimal positions.

Consider the following examples:

Example:

Solve the following decimals using direct division method by moving decimal positions:

(i) 0.55 ÷ 0.05 (ii) 0.125 ÷ 0.5 (iii) 2.25 ÷ 0.3

Solutions:

(i) 0.55 ÷ 0.05 = 55 ÷ 5 = **11**
(Both the decimals are with two decimal places).
So we get whole number.

$$\frac{\cancel{0.55}^{11}}{\cancel{0.05}_{1}} = 11$$

$$\begin{array}{r} 11 \\ 5\overline{)\ 55} \\ -\ 55 \\ \hline 0 \end{array}$$

(ii) 0. 125 ÷ 0.5 = 1.25÷ 5
3 decimal places – 1 decimal place
= 2 decimal places
= **0.25**

$$\frac{\cancel{.125}^{25}}{\cancel{.5}_{1}} = 0.25$$

$$\begin{array}{r} 0.25 \\ 5\overline{)1.25} \\ -\ 10 \\ \hline 25 \\ -\ 25 \\ \hline 0 \end{array}$$

(iii) 2.25 ÷ 0.3 = 22.5 ÷ 3 = 7.5
4 decimal places – 1 decimal place
= 3 decimal places

$$\frac{2.25}{0.3} = \frac{22.5}{3}$$

$$= 7.5$$

EXERCISE 4.8

A. Solve by converting decimals to fractions:

(1) 2.16 ÷ 0.6	(2) 5.76 ÷ 0.24	(3) 4.41 ÷ 0.3
(4) 7.84 ÷ 0.07	(5) 0.6017 ÷ 1.1	(6) 4.905 ÷ 4.5
(7) 7.84 ÷ 0.14	(8) 78.4 ÷ 0.7	(9) 10.24 ÷ 0.08
(10) 3.5308 ÷ 0.13	(11) 97.578 ÷ 0.039	(12) 10.26 ÷ 0.18

B. Solve the following decimals using direct division method by moving decimal positions:

(1) 0.016 ÷ 0.2	(2) 0.18 ÷ 0.3	(3) 0.36 ÷ 0.6
(4) 0.072 ÷ 0.8	(5) 0.121 ÷ 1.1	(6) 0.0169 ÷ 1.3
(7) 0.96 ÷ 0.4	(8) 9.018 ÷ 0.9	(9) 0.036 ÷ 1.2
(10) 0.0289 ÷ 1.7	(11) 0.0144 ÷ 1.2	(12) 0.072 ÷ 0.12
(13) 4.096 ÷ 0.16	(14) 0.325 ÷ 0.25	(15) 0.0196 ÷ 1.4

4.1.12 Use division to change fractions into decimals

Consider the following examples:

Examples: Change the following fractions into decimals:

(i) $\frac{2}{5}$ (ii) $\frac{10}{6}$

Solution: (i) $\frac{2}{5} = 2 \div 5$

$$\begin{array}{r|l} & \mathbf{0.4} \\ \hline 5 & \mathbf{20} \\ & -\mathbf{20} \\ \hline & \mathbf{0} \end{array}$$

So, $= \frac{2}{5} = \mathbf{0.4}$

(ii) $\frac{10}{6} = 10 \div 6$

$$\begin{array}{r|l} & \mathbf{1.66} \\ \hline 6 & \mathbf{1000} \\ & -\mathbf{6} \\ \hline & \mathbf{40} \\ & -\mathbf{36} \\ \hline & \mathbf{40} \\ & -\mathbf{36} \\ \hline \text{Remainder} \rightarrow & \mathbf{04} \end{array}$$

So, $\frac{10}{6} = \mathbf{1.66}$

EXERCISE 4.9

Change the following fractions into decimals:

(1) $\frac{5}{4}$	(2) $\frac{5}{3}$	(3) $\frac{4}{5}$	(4) $\frac{7}{10}$	(5) $\frac{15}{7}$
(6) $\frac{14}{9}$	(7) $\frac{5}{8}$	(8) $\frac{23}{9}$	(9) $\frac{32}{7}$	(10) $\frac{45}{13}$
(11) $\frac{50}{15}$	(12) $\frac{9}{14}$	(13) $\frac{11}{16}$	(14) $\frac{5}{12}$	(15) $\frac{17}{20}$
(16) $\frac{125}{60}$	(17) $\frac{245}{26}$	(18) $\frac{250}{8}$	(19) $\frac{300}{250}$	(20) $\frac{23}{25}$

4.1.13 Simplify decimal expressions involving brackets (applying one or more basic operations)

Simplifying decimals is a common activity. Just need some care of positions of decimals. Decimal expressions involving multiple operations and brackets are simplified using the rules of BODMAS.

Example 1: Simplify: 24.24 – (5.6 + 20.25 – 4.45)

Solution: First solve the expressions in the bracket.

24.24 – (5.6 + 20.25 – 4.45)
= 24.24 – (25.85 – 4.45)
= 24.24 – 21.40
= 2.84

(Operations of addition are performed, then subtraction in operations are performed)

Example 2. Simplify: 52.05 + (28.22 – 22.6) + (13.15 – 6.56)

Solution:

52.05 + (28.22 – 22.6) + (13.15 – 6.56) = 52.05 + 5.62 + 6.59
= 52.05 + 12.21
= 64.26

EXERCISE 4.10

Simplify:

(1) $5.6 + 7.22 - (2.24 + 4.68)$
(2) $(20.14 \times 5.6) + 10.9335$
(3) $4.6 + 6.07 + (23.35 - 8.30 + 8.34)$
(4) $14.3 - 2.8 + (1.84 + 3.29)$
(5) $(5.24 + 2.02) - 0.96 - 7.45 + (9.405 - 2.24)$
(6) $5.6 \times (25.5 - 12.2) + (2.3 + 2.6)$
(7) $45.234 + (18.024 - 6.66) - (0.457 + 9.945)$
(8) $3.45 \times 8.56 - 4.23 + (2.2 - 1.12)$
(9) $(6.6 \times 3.59) - (1.12 + 0.1) - 1.02$
(10) $(230.24 + 23.028) - 72.72 \div (6.42 + (14.045 - 6.3)$

4.1.14 Round off decimals up to specific number of decimal places

Consider the following examples.

Example 1.

Round off the following decimals to the nearest whole number:

(i) 4.2 (ii) 4.5 (iii) 4.7

This is a number line, 3.8, 3.9, 4.0, 4.1, 4.2, 4.3, 4.4, 4.5, 4.6, 4.7, 4.8, 4.9, 5.0, 5.1 and 5.2 are represented on it.

Solutions:

(i) **4.2**
Look at the number line.
4.2 is closer to 4 than to 5; and 4.2 is nearest to 4.
So, 4.2 = 4.

(ii) 4.5

Look at the number line.
4.5 is equally closer to 4 and 5.
If it is = or > 5, round of the numbers accordingly.
So, 4.5 becomes 5 after rounded off.

(iii) 4.7

Look at the number line.
4.7 is closer to 5 than to 4; and 4.6 is nearest to 5.
So, 4.7 = 5.

Rule: In order to round off a decimal nearest to the whole number. (i) Check the first decimal place or tenths digit to see if it is less or greater than 5. (ii) If it is < 5, it will remain unchanged. (iii) If it is = or > 5, rounded off the numbers accordingly.

Example 2.

Round off the following decimal numbers up to the one decimal places:

(i) 24.33 (ii) 50.67 (iii) 50.94

Solutions:

(i) 24.33

24.33 is closer to 24.3 than to 24.4; and 24.33 is nearest to 24.3
So, 24.33 after rounded of up to one decimal place = 24.3

(ii) 50.67

50.67 is closer to 50.7 than to 50.6; and 50.67 is nearest to 50.7
So, 50.67 after rounded off up to one decimal place = 50.7

(iii) 50.94

50.94 is closer to 50.9 than to 51 and 50.94 is nearest to 50.9
So, 50.94 after rounded off up to one decimal place = 50.9

Example 3.

Round off the following decimal numbers up to the two decimal places:

(i) 9.354 (ii) 58.687

Solutions:

(i) 9.354

9.354 is closer to 9.35 than to 9.36; and 9.354 is nearest to 9.35

So, 9.354 after rounded off up to two decimal places = 9.35

(ii) 58.687

58.687 is closer to 58.69 than to 58.68; and 58.687 is nearest to 58.69

So, 58.687 after rounded off up to two decimal places = 58.69

EXERCISE 4.11

1. Round off the following decimals as whole numbers.

(1) 2.3	(2) 5.6	(3) 7.7	(4) 6.6
(5) 9.9	(6) 8.3	(7) 7.8	(8) 50.2
(9) 58.6	(10) 78.2	(11) 81.7	(12) 99.9

2. Round off the following decimals up to one decimal place.

(1) 32.38	(2) 25.156	(3) 6.17	(4) 6.42
(5) 76.798	(6) 95.24	(7) 12.86	(8) 5.95
(9) 3.432	(10) 11.7681	(11) 50.4752	(12) 60.1536

3. Round off the following decimals up to two decimal places.

(1) 32.386	(2) 25.056	(3) 6.775	(4) 6.422
(5) 76.798	(6) 8.4832	(7) 0.9627	(8) 58.1905
(9) 4.0098	(10) 40.9807	(11) 70.4908	(12) 19.0185

4.1.15 Convert fractions to decimals and vice versa

Consider the following examples:

Example 1. Convert the fraction $\frac{6}{10}$ into decimals.

Solution: $\frac{6}{10}$ = 6 tenths = 0.6

Example 2. Convert the fraction $\frac{2}{5}$ into decimals.

Solution:

$\frac{2}{5} = \frac{2 \times 2}{5 \times 2} = \frac{4}{10}$ (Multiply 2 by 2 and 5 by 2, we get the new fraction).

= 4 tenths = 0.4

Example 3. Convert the fraction $\frac{45}{1000}$ into decimals.

Solution: $\frac{45}{1000}$ = 45 Thousandths = 0.045

Let us convert the decimals 0.68 to a fraction

$$0.68 = \frac{0.68}{1\,0\,0} = \frac{68}{100} = \frac{\cancel{68}^{17}}{\cancel{100}_{25}} = \frac{17}{25}$$

Remember: For the numerator, we write the given number without the decimal point. For the denominator, we write 1 for the decimal point and put zeros after 1 according to the decimal places. Then simplify if possible.

Example 4: **Convert following decimals into fractions in their lowest form.**

(i) 0.8 (ii) 0.625

Solution:

(i) $0.8 = \frac{0.8 \times 10}{10} = \frac{8}{10}$

$= \frac{\cancel{8}^{4}}{\cancel{10}_{5}} = \frac{4}{5}$

Thus $0.8 = \frac{4}{5}$

(ii) $0.625 = \frac{0.625 \times 1000}{1000}$

$= \frac{\cancel{625}^{\cancel{125}^{\cancel{25}^{5}}}}{\cancel{1000}_{\cancel{200}_{\cancel{40}_{8}}}} = \frac{5}{8}$

Thus $0.625 = \frac{5}{8}$

EXERCISE 4.12

A. Convert the following fractions into decimals:

(1) $\frac{7}{8}$ (2) $\frac{9}{10}$ (3) $\frac{2}{5}$ (4) $\frac{17}{25}$ (5) $\frac{61}{50}$

(6) $\frac{19}{20}$ (7) $\frac{49}{40}$ (8) $\frac{71}{80}$ (9) $\frac{451}{500}$ (10) $\frac{79}{250}$

(11) $\frac{83}{100}$ (12) $\frac{111}{400}$ (13) $\frac{777}{800}$ (14) $\frac{551}{1000}$ (15) $\frac{999}{10000}$

B. Convert the following decimals into fractions:

(1) 0.5 (2) 1.05 (3) 3.56 (4) 0.565

(5) 0.023 (6) 0.25 (7) 0.345 (8) 35.506

(9) 0.064 (10) 0.945 (11) 41.625 (12) 46.1024

4.1.16 Solve real life problems involving decimals

We have already learnt four operations on decimals. Now we will learn real life problems based on operations.

Example 1. Ahmed and Ali are two brothers, they saved Rs 1245.50 and Rs 1050.50 respectively. How much amount they saved altogether?

Solution:

$$\begin{array}{r} 1245.50 \\ +\ 1050.50 \\ \hline 2296.00 \\ \hline \end{array}$$

Ahmed saved money = Rs 1245.50
Ali saved money = Rs 1050.50
Both brothers saved = Rs 1245.50 + Rs 1050.50
= Rs 2296.00

Example 2. Price of 15 pencils is Rs 97.50. Find the price of one pencil.

Solution:

The price of 15 pencils is Rs 97.50

Therefore, the price of 1 pencil $= 97.50 \div 15$

$$= \frac{9750}{100} \div 15$$

$$= \frac{9750}{100} \times \frac{1}{15}$$

$$= \frac{\overset{65}{\cancel{975}}}{10 \times \underset{1}{\cancel{15}}} = \frac{65}{10}$$

$$= 6.50$$

Thus, price of one pencil is Rs 6.50

EXERCISE 4.13

(1) Sahir bought two chickens of weight 1.450 kg and other of 1.685 kg respectively. Find the total weight of both chicken.

(2) A general store sold 200.750 kg of flour and 98.500 kg of sugar. How much more flour was sold than sugar?

(3) Saira bought a necklace of gold of mass 2.565 g and bangles of mass 8.875 g. What is the total mass of these two items?

(4) Aftab saved Rs 3206.75 from monthly salary of Rs 35200. Find his monthly expenditure.

(5) Noor communications earn Rs 2345.75 in a day by selling easy loading balance. How much amount he will earn in one month?

(6) A bundle contains 119.5 m of wire. If one piece of wire measuring 29.92 m is sold. Find the length of remaining piece of wire.

(7) The height of a pole is 18.75 m. We have ladder of $\frac{1}{3}$ of pole. How much length is left out?

(8) A school uniform is made from 4.5 m cloth. How many uniforms can be made from 31.5 m cloth?

(9) A lady used 3.60 kg of salt in preparing different kinds of food for her family in the month of April. How much salt is used per day?

(10) The price of 2.5 kg ghee is Rs 391.25. Find the price of 1 kg.

4.2 PERCENTAGES

4.2.1 Recognize percentage as a special kind of fraction

The word "Percent" is formed of two words, "Per" and "Cent". "Per" means out of and "Cent" means hundred. Therefore, the word PERCENT is used to mean one hundred, and it is represented by the symbol%.

1% makes $\frac{1}{100}$ means 1 out of 100 = 0.01

10 % makes $\frac{10}{100}$ means 10 out of 100 = 0.1

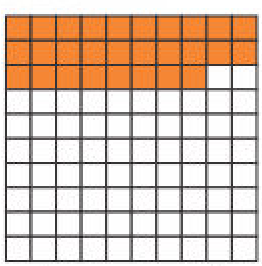
and 100% makes $\frac{100}{100}$ means 100 out of 100 = 1

Thus, percentage is a special kind of fraction.

Activity

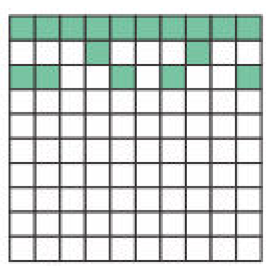
Express the shaded parts as percentages.

(i)

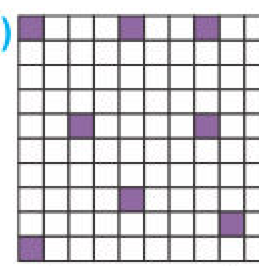


$\frac{28}{100}$ = 28%

(ii)



(iii)



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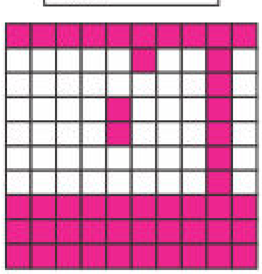

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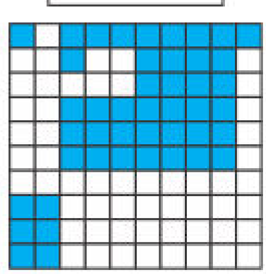

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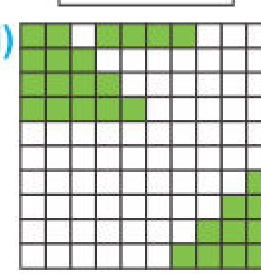

Teacher's Note Teacher should explain the concept of percentage as a kind of fraction.